A Modular Pipeline for Inferring Planetary Companions from Gaia DR3 RUWE

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INTRODUCTION

The *European Space Agency's Gaia mission* measured stellar positions with tens of microarcsecond precision, enabling detection of subtle astrometric perturbations caused by unresolved companions.

Gaia fits a **5-parameter single-star astrometric model** to each source: **Right Ascension (RA), Declination (Dec), Parallax, Proper motion in RA, Proper motion in Dec**

If a star hosts a planetary companion, the photocenter of the system undergoes orbital motion that is not included in the standard 5-parameter model. This produces excess residuals in Gaia's along-scan measurements. And, those residuals are compressed into a single diagnostic quantity: **RUWE - Renormalized Unit Weight Error**.

Scientific Goal

Given Gaia's published RUWE, infer what planetary configurations could reproduce the observed astrometric excess

GAIA RUWE

Chi-square statistic goodness-of-fit:

$$X^2 = \Sigma \left(\frac{r_i}{\sigma_i}\right)$$

r_i : along-scan residuals
σ_i : uncertainties

$$RUWE = \sqrt{\frac{X^2}{(N-5)}}$$

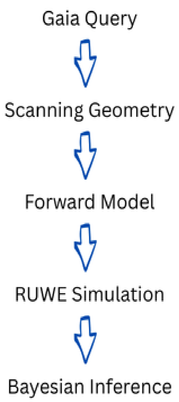
N : number of along-scan obs
5 : number of fitted parameters

- RUWE ≈ 1.0 → Well-fit single-star model
- RUWE > 1.0 → Possible unresolved companion

PIPELINE ARCHITECTURE

Gaia Query

We retrieve the following quantities from Gaia DR3: **RA, Dec, Parallax, Proper Motions, Photometric magnitudes, RUWE, Observation times, Scan Angles, AL Parallax Factors.** These are required to reproduce Gaia's geometry.



Forward Modeling

Single-star Model

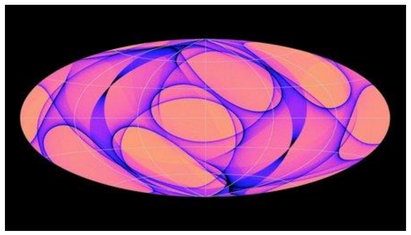
$$AL(t) = \Delta\alpha(t)\,\sin(\psi(t)) + \Delta\delta(t)\,\cos(\psi(t))$$

$\Delta\alpha,\ \Delta\delta$: proper motion
$\psi(t)$: scan angles

Planet Model

Inputs: semi-major axis (a), inclination (i), eccentricity (e), longitude of ascending node (Ω), argument of periastron (ω), time of periastron, planet mass, star mass, time series

Output: Gaia Along-Scan (AL) positions



Process:
- Compute **photocenter motion** using **Thiele-Innes formalism**
- **Project total position** onto Gaia's AL

RUWE Simulation

To reproduce Gaia's fitting procedure, we construct the design matrix:

$$A = [\sin\psi \quad \cos\psi \quad \pi_{AL} \quad t\sin\psi \quad t\cos\psi]$$

We compute residuals after fitting the 5-parameter model:

$$r = AL - A(A^TA)^{-1}A^TAL$$

The Modeled RUWE is:

$$RUWE_{model} = \sqrt{\frac{\Sigma\left(\frac{r}{\sigma}\right)^2}{N-5}}$$

Bayesian Inference

We use **PyMC** to sample the posterior. We frame the problem probabilistically:

$$P(\theta|RUWE_{obs}) \propto P(RUWE_{obs}|\theta)\,P(\theta)$$

$\theta : \{a, i, e, \Omega, \omega, M_p\}$

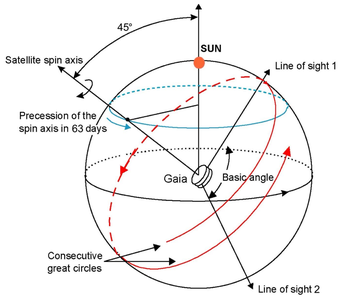
Our goal is to determine which planetary configurations are consistent with the observed Gaia RUWE

Likelihood Function: Compare the Gaia-published RUWE to the simulated RUWE

$$RUWE_{obs} \sim \mathcal{N}(RUWE_{model}, \sigma_{RUWE})$$

Output: Posterior samples of planet & system parameters

This transforms RUWE from a diagnostic flag into a quantitative inference tool.



NEXT STEPS

- **Matching scan geometry to observation epochs**
- **Consistency in astrometric units**
- **Test each part of the pipeline and implement compatibility changes**
- **PyTensor compatible orbital model**
- **Validation on test cases**
- **Along-scan noise model implementation**

REFERENCES

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ACKNOWLEDGEMENTS

Advisor: Prof. Jiayin Dong
Tanay Agrawal, Johnny Cheng, Grace Houlihan, Naavya Jain, Noori Khurshid, Tanmay Raswant, Zixuan (Jay) Yan

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